

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{\ln 2}{5730} = 1.2097 \times 10^{-4} \text{ [yr}^{-1}\text{] or } 3.84 \times 10^{-12} \text{ [s}^{-1}\text{]} \text{ (1)}$ Correct use of 70% i.e. 0.7 (1) Taking logs (1) so $t = \frac{1}{\lambda} \ln \left(\frac{N_0}{N} \right) = \frac{1}{1.2097 \times 10^{-4}} \ln \left(\frac{1}{0.7} \right) = 2948 \text{ [yr] or } 9.29 \times 10^{10} \text{ [s]} \text{ (1)}$ Award 3 marks for 9950 years Alternative: Correct use of 70% i.e. 0.7 (1) Substitution e.g. $0.7 = \frac{1}{2^n}$ (1) Taking logs e.g. $\ln 0.7 = -n \ln 2$ (1) Correct answer (1)	1 1	1		4	4	
	(b)		Substitution (1) $\left(\frac{A}{A_0} \right) = e^{- (1.2097 \times 10^{-4}) (7000)}$ = 0.429 (1) = 42.9% (1) Alternative: Number half-lives = $\frac{7000}{5730} = (1.22) \text{ (1)}$ Substitution i.e. $A = \frac{A_0}{2^{1.22}} \text{ (1)}$ Answer = 43% (1)	1	1 1		3	3	
			Question 6 total	3	4	0	7	7	0